Solen

The Settlement Layer for the Modular Future

“Built from scratch in Rust for the modular blockchain era.”

solenchain.io

1. WHAT IS SOLEN

Solen is a new Layer 1 blockchain built from scratch in Rust, designed as the settlement foundation for the modular blockchain era. Rather than trying to do everything on a single chain, Solen serves as the secure, high-performance base layer where rollups settle proofs, assets are bridged across domains, and users interact through intelligent accounts that abstract away the complexity of crypto.

Every component of Solen was purpose-built. There are no forks of existing chains, no inherited technical debt, and no design compromises carried over from earlier generations. The consensus engine, execution environment, virtual machine, peer-to-peer networking, storage layer, and smart account system were all written from the ground up to work together as a unified, cohesive stack.

Solen runs a BFT proof-of-stake consensus protocol with stake-weighted proposer selection, producing blocks every 2 seconds with deterministic finality at epoch boundaries. A built-in WASM virtual machine executes smart contracts with fuel metering and deterministic guarantees. On-chain governance allows the community to propose and vote on protocol parameter changes, including block time, minimum validator stake, and fee burn rates, without hard forks or centralized intervention.

The result is a blockchain that is fast, secure, and genuinely decentralized from day one, with the developer experience and end-user safety features that the industry has been waiting for.

> *“Fast, secure, and genuinely decentralized from day one.”*

2. THE PROBLEM & OPPORTUNITY

The blockchain industry has a settlement problem. As the ecosystem moves toward a modular architecture, where execution happens on rollups and app-specific chains, the need for a reliable, purpose-built settlement layer has never been greater. Yet today's options are fundamentally inadequate.

Ethereum, the current default for rollup settlement, was designed over a decade ago as a general-purpose smart contract platform. It processes roughly 15 transactions per second on the base layer, with gas fees that routinely spike to $20-$50 or more during periods of network congestion. Settlement finality takes 12+ minutes. Its account model offers no native recovery mechanism: lose your private key and your assets are gone permanently. These are not bugs; they are architectural constraints baked into a system that was never designed for the role it is being asked to play.

Alternative Layer 1s have optimized for raw throughput but sacrifice decentralization, lack native rollup support, or bolt on account abstraction as an afterthought rather than a core protocol feature. None were designed from the ground up as settlement infrastructure for a modular world.The opportunity is massive. The total value locked in rollups exceeded $40 billion in 2025 and continues to grow exponentially. Hundreds of new rollups are launching each year, spanning gaming chains, DeFi chains, social chains, and enterprise chains, and every one of them needs a settlement layer they can trust. The market for modular blockchain infrastructure is projected to reach tens of billions of dollars, and the settlement layer sits at the most critical chokepoint in the entire stack.

Solen is built to capture that opportunity, not by competing with rollups for execution, but by providing the secure, fast, and affordable foundation they all depend on.

> *“The settlement layer sits at the most critical chokepoint in the modular stack.”*

3. THE SOLUTION

Solen addresses the settlement gap with five core innovations that work together as an integrated system:

Smart Accounts

Every account on Solen is a smart account by default. Users get programmable authentication, including Ed25519, passkeys and biometrics, and threshold multisig; social recovery through trusted guardians; session keys that let applications act on a user's behalf with strict spending and time limits; and the ability to swap authentication methods without changing an account address. Wallet recovery, biometric login, and granular permissions become protocol-level features, not third-party add-ons that introduce new trust assumptions.

Native Rollup Settlement

Solen provides first-class support for both optimistic and ZK rollup settlement at the protocol level. Rollups register on-chain, submit state commitments, and have their proofs verified natively. There are no custom bridge contracts, fragmented liquidity pools, or middleware dependencies. Cross-rollup asset movement is handled through protocol-secured bridge vaults, enabling seamless interoperability between execution domains.

Intent-Based Execution

Instead of requiring users to construct exact transaction parameters, Solen's native intent system lets users declare what they want to accomplish: swap token A for the best price, bridge assets to Chain X, or pay up to Y for an NFT. Specialized solvers compete to fulfill these intents optimally, giving users better execution without requiring technical fluency. The UX shifts from execute this specific call to achieve this outcome.

High-Performance Consensus

Solen's BFT proof-of-stake consensus produces blocks every 2 seconds with deterministic finality at epoch boundaries backed by 2/3+ stake-weighted attestations. Proposer selection uses a verifiable randomized algorithm weighted by stake, preventing gaming and ensuring fair block production. Validators are held accountable through automatic on-chain slashing for double-signing, downtime, and invalid state proposals, with cryptographic proof requirements that prevent false accusations.

WASM Smart Contracts

A purpose-built WebAssembly virtual machine executes smart contracts with fuel-based gas metering, cranelift compilation, and NaN canonicalization for deterministic cross-platform execution. Developers write contracts in Rust using Solen's contract SDK, with on-chain source verification so anyone can audit deployed code. The SRC-20 token standard provides ERC-20 equivalent functionality with native support for minting, transfers, allowances, and metadata.

> *“The UX shifts from execute this specific call to achieve this outcome.”*

4. PRODUCT & BUSINESS MODEL

Product Stack

Solen is not just a blockchain node. It is a complete ecosystem of developer and user-facing tools:

- Solen Node - Full validator node software written in Rust, handling consensus, execution, peer-to-peer networking, and RPC serving in a single optimized binary
- Solen CLI - Command-line interface for key management, transfers, staking, governance voting, contract deployment, and validator operations
- Solen Wallet - Cross-platform desktop wallet for Windows, macOS, and Linux built with Tauri, featuring encrypted key storage, multi-account management, Base58 addresses, and one-click account backup
- SolenScan - Full-featured block explorer with dark and light mode, real-time block and transaction tracking, validator monitoring, contract source verification, governance proposal display, and token analytics
- Solen Documentation - Comprehensive developer docs covering architecture, APIs, CLI reference, smart contract SDK, deployment guides, and tokenomics
- Contract SDK - Rust SDK for writing WASM smart contracts with storage, events, and system call abstractions

Every component is open source under MIT/Apache-2.0 dual license.

Economic Model

Solen's economic engine is built around the native SOLEN token with a fixed supply of 2 billion tokens and no inflation. Value accrues through:

- Transaction fees - Every operation pays a gas fee denominated in SOLEN. A governance-configurable percentage of fees is permanently burned, creating deflationary pressure as network usage grows.
- Settlement fees - Rollups settling on Solen pay fees for state commitment submissions and proof verification, creating recurring demand tied directly to ecosystem growth.
- Staking yield - Validators and delegators earn rewards from block production, incentivizing network security while distributing value to participants.
- Account creation surcharge - New account creation carries a modest gas surcharge that prevents state bloat while generating additional fee revenue.

The model is designed so that as more rollups settle on Solen and more users transact, the deflationary burn increases, staking demand grows, and token value compounds, creating a virtuous economic flywheel.

5. GO-TO-MARKET STRATEGY

Phase 1: Developer Foundation (Current)

Solen's initial go-to-market focuses on building a robust developer community and proving the technology on a public testnet. The testnet is live today with active validators, deployed contracts, and governance proposals being executed on-chain. Developer documentation covers the full stack from node setup to smart contract deployment. All tooling, including wallet, explorer, CLI, and faucet, is production-ready and publicly accessible.

Phase 2: Rollup Partnerships

The primary growth driver is rollup adoption. Solen will partner with rollup frameworks such as OP Stack, Arbitrum Orbit, Sovereign SDK, and custom rollup kits to offer Solen as a settlement option. Native rollup registration, proof verification, and bridge vault infrastructure make integration straightforward compared with building custom settlement contracts on general-purpose chains. Each rollup that settles on Solen brings its entire user base into the ecosystem.

Phase 3: Application Ecosystem

Smart accounts and intent execution are Solen's differentiated user-facing features. By partnering with wallet providers, DeFi protocols, and consumer applications that want to offer passkey login, social recovery, and intent-based UX, Solen can drive end-user adoption through the applications people actually use, without requiring users to understand the underlying blockchain.

Community Building

- Ecosystem grants for developers building on Solen
- Hackathons and bounty programs targeting Rust and blockchain developers
- Validator onboarding program with documentation and support
- Ambassador program for community-driven growth in key markets

Phase 4: Enterprise & Institutional

Solen's governance system, configurable parameters, and native rollup support make it well-suited for enterprise use cases, including supply chain, financial settlement, identity, and compliance-focused applications that need their own execution environment but want the security guarantees of a decentralized settlement layer.

6. EARLY MOMENTUM

Solen has achieved significant technical milestones in a compressed timeframe:

- Complete L1 implementation - Consensus engine, execution environment, WASM VM, peer-to-peer networking, storage, RPC, mempool, and governance are all fully implemented and running in production on testnet
- Public testnet live - Multiple validators are producing blocks with real BFT consensus, attestation-based finality, and epoch transitions operating continuously
- Security hardened - Four comprehensive adversarial security audit passes completed, including cross-chain vulnerability analysis drawing from Ethereum, Cosmos, and Solana incident patterns. More than 20 vulnerabilities were identified and fixed across consensus, execution, and networking layers
- Full tooling suite shipped - Desktop wallet with cross-platform CI/CD builds, block explorer with dark mode and contract verification, CLI with 20+ commands, faucet with rate limiting, and comprehensive documentation
- On-chain governance tested - Governance proposals have been submitted, voted on, finalized, and executed on the live testnet, including parameter changes to block time and minimum validator stake
- Smart contract ecosystem - SRC-20 token standard deployed and verified, with contract SDK and developer documentation ready for third-party builders
- Preparing for external audit - Codebase is ready for third-party security review, with a detailed audit briefing document prepared for professional auditors

7. FUNDING & VISION

The Ask

Solen is raising a seed round to accelerate the path from testnet to mainnet and fund the first wave of ecosystem growth. Capital will be deployed across four priorities:

- Mainnet launch and infrastructure - Production node hosting, monitoring, and redundancy for launch validators; security audit engagement with a top-tier blockchain auditing firm
- Ecosystem development - Developer grants, hackathon sponsorships, and bounty programs to seed the initial application ecosystem on Solen
- Team expansion - Hiring additional protocol engineers, developer relations, and community management to support growth
- Partnerships and integration - Rollup framework integrations, wallet provider partnerships, and enterprise pilot programs

The Vision

Solen's long-term vision is to become the default settlement layer for the modular blockchain ecosystem, the foundational infrastructure that rollups, applications, and institutions trust to secure their most critical operations. Just as cloud computing coalesced around a few trusted infrastructure providers, the modular blockchain stack will coalesce around the settlement layers that offer the best combination of security, performance, developer experience, and economic alignment.

Solen is building that layer from scratch, with no shortcuts, and with the conviction that the settlement layer of the future should be purpose-built for that role from day one.

Mainnet target: Q2 2026

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github.com/Solen-Blockchain

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